

Virtual Juggling: A Look at HyperThreading

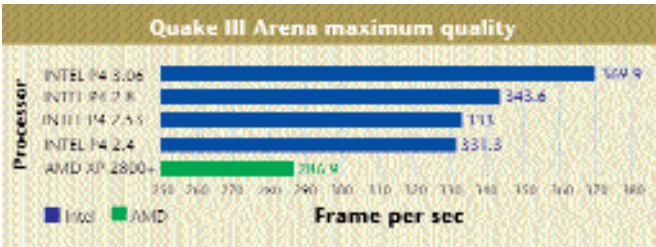
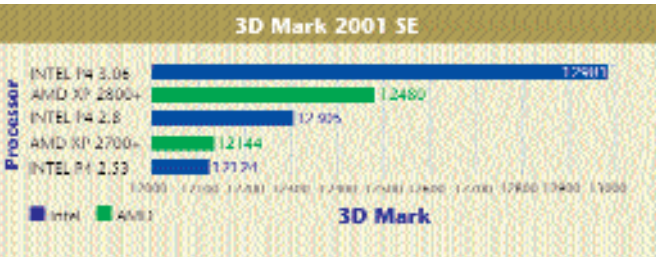
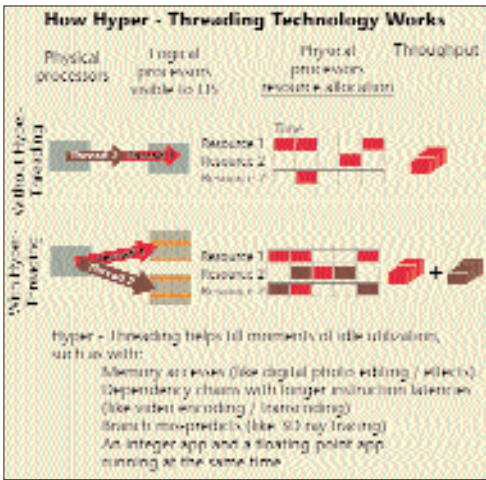
With the launch of their 3.06 GHz HyperThreaded processors, Intel claims to have delivered two processors in a single package. So what does HyperThreading (HT) do? Faster clock speeds is one way of increasing the computing power and the other way of increasing the output is by doing maximum work in a single clock cycle.

Consider a juggler who can theoretically juggle four balls but is juggling only three. HT is a means of ensuring that the number of balls is never less than four, an efficiency-increasing prospect. Under a multi-tasking environment, this means that the system resources are used in a manner such that at the end, a specific process is completed in less time. Technically, HT entails sending two data threads (think balls that need to be juggled) simultaneously to the processor (non-HT processors can handle only one thread at a time). The operating system gets an impression of two processors being present in the system and system resources are allocated in such a way that both the threads are executed simultaneously, allowing them to complete at the same time, instead of having to wait for one process to get

over first. This increases efficiency and results in a decent performance gain.

So much for the theory, let us talk real-world now. A performance gain of not more than 25 per cent is expected from this technique, vis-à-vis traditional processors. Some tasks will be easier to carry out while others may suffer. In general, if one of the threads from a multi-threaded application (or one thread from one of two multitasked applications) is processor-intensive then HT will not have a positive impact on performance. Benefits are promised for two threads that do not strain resources. If multiple threads need access to the same resources, then the system could be rendered unstable or at the least, performance would take a nose dive: therefore, encoding music while watching a movie might yet pose a challenge to an HT processor.

HyperThreading requires certain prerequisites: the motherboard should support it, a small challenge, and more importantly software must be written with multiple threads in mind. Intel will supply a relevant compiler that will help developers do the latter. The OS should also be multi-threaded—Linux and Windows XP Pro are.



more than its nearest AMD rival, the Athlon 2800+ (12480 3D Marks). *Quake III Arena* is more comfortable with a processor that gives it raw speeds as is evident by the 369.9 fps it gifts the P4 3.06 GHz (at maximum quality)—way more than what the XP 2800+ scores (286.9 fps). Although the primary reason is the speed advantage, the huge L2 cache of the P4 processor (512 KB as against the 256 KB that the 2800+ employs) is another reason for the P4's domination in gaming. Thus, you will find that the first four positions in the Gaming and Entertainment STTI score table are held by Intel processors, the Intel Pentium 4 2.4 GHz coming in at number four with an 83.94 STTI, while the P4 3.06 GHz takes pole with a score of

94.32 STTI. A cheaper Intel solution would thus revolve around the P4 2.4 GHz processor, which costs the least of the four at Rs 12,800.

The closest AMD competitor to the 3.06 GHz Intel chip, the AMD XP 2800+ scores 81.80 STTI at a price of Rs 23,820. Going for this chip makes little sense (unless you already have an AMD-based motherboard) since it performs less than the P4 2.4 GHz but costs Rs 11,020 more than its rival.

However, to spot a bargain in the making, we need to climb down a couple of rungs to the 2400+ processor. From here to the 1800+, the AMD chips are a great mix of performance and affordability. The Palomino-cored AMD XP 1800+ shines with a score of 73.47 STTI and a price of Rs 4,140. Compare this to the nearest Intel counterpart, the P4 2.2 GHz (scoring 73.19 STTI) which is priced more than twice as much at Rs 11,600, and it becomes evident that AMD chips still rule the roost when it comes to performance at a digestible price.

Our recommended solution

These components make a killer combination for the ultimate gaming and entertainment solution. The MSI 845PE MAX2 offers everything a hardcore gamer could ask for and more. We tested the board using DDR 266 memory and came up with some amazing benchmark results. However, the real performance would be noticed when this board is coupled with DDR 333 memory and a fast Intel processor.

The MSI offers six-channel audio, onboard RAID, core frequency and voltage adjustments, and six PCI slots, leaving enough room for expansion. The board is also Bluetooth ready.